

Enhancing Online Video Recommendation via a Coarse-to-fine Dynamic Uplift Modeling Framework

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Abstract

The popularity of short video applications has brought new opportunities and challenges to video recommendation. In addition to the traditional ranking-based pipeline, industrial solutions usually introduce additional distribution management components to guarantee a diverse and content-rich user experience. However, existing solutions are either non-personalized or fail to generalize well to the ever-changing user preferences. Inspired by the success of uplift modeling in online marketing, we attempt to implement uplift modeling in the video recommendation scenario to mitigate the problems. However, we face two main challenges when migrating the technique: 1) the complex-response causal relation in distribution management problem, and 2) the modeling of long-term and real-time user preferences. To address these challenges, we correspond each treatment to a specific adjustment of the distribution over video types, then propose a Coarse-to-fine Dynamic Uplift Modeling (CDUM) framework for real-time video recommendation scenarios. Specifically, CDUM consists of two modules, a coarse-grained module that utilizes the offline features of users to model their long-term preferences, and a fine-grained module that leverages online real-time contextual features and request-level candidates to model users' real-time interests. These two modules collaboratively and dynamically identify and target specific user groups, and then apply treatments effectively. We conduct comprehensive experiments on two offline public datasets, an industrial offline dataset, and an online A/B test, demonstrating the superiority and effectiveness of CDUM. The proposed method is fully deployed on Kuaishou platform, serving hundreds of millions of users every day. Our code and datasets are available at https://github.com/UpliftVideo/CDUM.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Uplift Modeling; Coarse-to-fine; Video Recommendation

ACM Reference Format:
Chang Meng, Chenhao Zhai, Xueliang Wang, Shuchang Liu, Xiaoqiang Feng, Lantao Hu, Xiu Li, Han Li, and Kun Gai. 2025. Enhancing Online Video Recommendation via a Coarse-to-fine Dynamic Uplift Modeling Framework. In *Proceedings of the Nineteenth ACM Conference on Recommender Systems (RecSys '25), September 22–26, 2025, Prague, Czech Republic.* ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3705328.3748070

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RecSys '25, Prague, Czech Republic

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ACM ISBN 979-8-4007-1364-4/25/09
https://doi.org/10.1145/3705328.3748070

1 INTRODUCTION

The widespread short video platforms as a new form of social media have significantly impacted everyone's lifestyle as well as the industry's landscape. Specifically, a recommender system is deployed to analyze the varying user preferences and recommend videos that meet the heterogeneous demands of users. While the

main recommendation process may adopt standard learning-to-rank methods [24, 38, 39, 44, 45] (which mainly rely on user behavior sequence modeling [8–10]) that rank and select videos that are most aligned with the users' preferences, industrial solutions often introduce extra decision-making modules that control the distribution of different video types (e.g., duration types, traffic boosters, and content categories) to guarantee a content-rich user experience and avoid biased recommendation. For example, a user may expect to see both short videos and long videos in the recommendation even when the ranking model happens to select merely long videos or merely short videos, so we need a strategy to ensure the presence of both types. Compared to common recommendation task formulations, the key difference of this type of problem is that it does not directly optimize the user feedback but aims to conduct reasonable intermediate "decisions" in the pipeline that control the overall impression of the recommendation.

Conventionally, the technical solution usually involves a module that engages "distribution management" over different video types and we have deployed various straightforward solutions in our industrial system: The general framework first statistically estimates the user's behavioral (e.g. watch time) distribution over different video types and conducts a heuristic-based management strategy according to the analytical results. Though empirically effective, these naive approaches either fail to model the personalized preference over video types or struggle to capture the dynamic causal relations between the distribution management decision and the user behavior statistics. In order to achieve better generalization performance, we need a robust cause-effect modeling framework that can effectively explore and adjust to user interests.

To address the aforementioned task, we find it feasible to reformulate the problem with uplift modeling, which is a personalized causal effect modeling technique that is widely used in online marketing scenarios like the distribution of coupons, discounts, and bonuses. The key functionality is to measure the incremental impact (i.e., uplift) of treatments or interventions for populations [7, 17, 22, 33] and it makes decisions based on the measurement. Representative methods include meta-learning models such as T-learner [19], S-learner [19], as well as tree-based models [2, 30, 47], and Deep Uplift Modeling (DUM) [17, 22, 23, 25, 35, 36, 48]. In this work, inspired by the success of uplift modeling in online marketing scenarios, we reformulate the distribution management problem in our video recommendation pipeline as a uplift modeling task.

Nevertheless, applying the uplift modeling strategy in the video recommendation scenario presents additional challenges:

- *Complex-response causal relation in distribution management problem*: as illustrated in Figure 1, in online marketing scenario, a binary treatment (e.g. exposure of a coupon) receives a direct user response, making the causal estimation relatively straightforward. Yet, in video recommendation, we aim to manage the (duration) type-level distribution for recommendations which is not directly reflected by the video-level user feedback. For example, increasing the exposure of short videos may boost commercial interactions while increasing the exposure of long videos may improve user's watch time. Intuitively, we need a reasonable uplift modeling solution that can fully express the distribution management space in the short video recommendation scenario.
- *Long term and real-time preference modeling*: In video recommendation scenarios, user preference modeling not only expands into long-term histories (i.e. over a week) but also focuses on real-time dynamic changes. While uplift modeling can apply statistical analysis on the user's long-term history, existing methods are incapable of capturing real-time dynamics of the user preference. As a result, we need a decision making strategy that can balance between the two granularities.

To address these two challenges, we carefully design a **C**oarse-to-fine **D**ynamic **U**plift **M**odeling framework for Real-time Video Recommendation (CDUM), which includes two modules: On the one hand, the Coarse-grained Preference Modeling (CPM) component adopts an offline update paradigm that captures the user's long-term preferences. To address the complex-response causal relation in distribution management problem challenge, it corresponds each treatment to a specific adjustment of the distribution over video types with carefully designed causal effect labels and cross-treatment model architecture. On the other hand, the Fine-grained Interest Capture (FIC) module uses online real-time contextual features and request-level candidates to capture users' real-time preferences. It constructs an multi-task learning module that adjusts the output scores of CPM and aligns the user's different preferences across video types during the online process.

In summary, our work makes the following contributions:

- We highlight the problem of distribution management in the video recommendation pipeline, and identify the difficulties and challenges faced in implementing uplift modeling, specifically, the design and utilization of treatments, as well as the decision making under users' long-term and real-time interests.
- We have innovatively proposed Coarse-to-fine Dynamic Uplift Modeling in the real-time video recommendation scenario. This approach combines offline features with online real-time features to model users' long-term preferences and real-time interests in a coarse-to-fine paradigm, dynamically identifying and targeting specific user groups and applying treatments effectively.
- We validate our model on two public datasets, an industrial dataset from Kuaishou platform, and an online A/B experiment on this platform. Our model not only demonstrates superior offline performance but also achieves significant improvements in consumption metrics and user retention on the platform. Our technique is ultimately deployed on the online platform across various stages in the recommendation pipeline, serving hundreds of millions of users.

2 RELATED WORK

Uplift models aim to identify the target user groups for each specific treatment by accurately estimating the individual treatment effects (ITE). Existing Uplift modeling approaches fall into three categories: meta-learner based methods, tree-based methods, and neural network-based methods.

Meta-learner based methods [19, 42] are a classic framework that focuses on using existing prediction methods to learn user responses. S-learner [19] and T-learner [19] are two commonly used meta-learning paradigms. The S-learner combines the treatment variable with other features and estimates ITE through a single model. The T-learner builds models separately for the control and

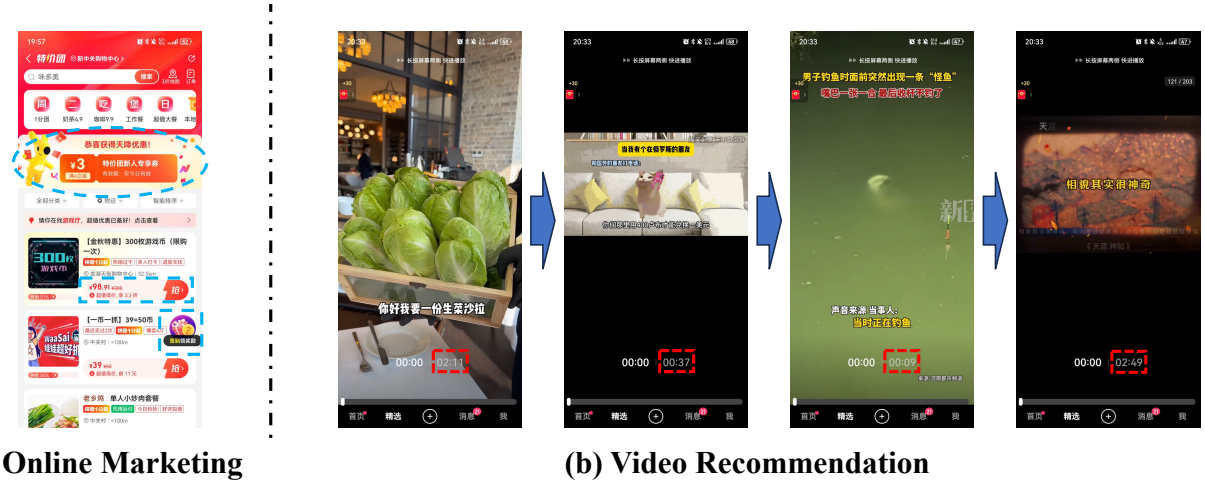


Figure 1: A example of the difference between online marketing and video recommendation scenarios.

treatment groups and estimates ITE through the subtraction between the two models. In both paradigms, when there is a significant imbalance in the amount of data between the treatment and control groups, this imbalance may significantly affect model performance, thereby affecting the estimation of ITE. To address this issue, the X-learner [19] method has been improved based on the T-learner. It introduces the concept of propensity scores and two additional ITE estimators, combining propensity scores to weight the estimation of ITE. This method performs particularly well when there is a significant difference in the number of samples between the treatment and control groups, or when the ITE pattern is relatively simple. Tree-based methods [2, 29, 30, 47] utilize specific tree or forest structures to progressively separate sensitive subgroups corresponding to each treatment from the entire population using different metric-based splitting criteria, with treatment information included in the calculations of the splitting process.

Due to the outstanding feature extraction capabilities of deep learning, methods based on neural networks [3, 25, 33, 34, 40] have become a popular framework for uplift estimation. They introduce more complex and flexible architectures to model the response process to treatment, enabling the learning of more accurate user responses or uplift estimation values. These methods are widely applied in industries such as online marketing and precision medicine [13, 14, 23, 49]. For instance, in online marketing, these methods utilize offline user features to perform uplift modeling, and select the target population that is sensitive to incentives (e.g. coupons, discounts, and bonuses) and have a positive effect based on the estimated ITE value (i.e. uplift), which reduces marketing costs while significantly improving indicators such as commission conversion rate. For example, EUEN [17] uses two sub-neural networks to estimate the conversion probability of the control group and the uplift effect respectively; DESCN [48] is an end-to-end multi-task cross-network designed to comprehensively capture the relationship between treatment and real response while alleviating treatment bias and sample imbalance issues; EFIN [22] focuses on explicit feature interaction awareness, especially the sensitivity of non-treatment features to specific treatments, and enhances the model's robustness through an intervention constraint module. However, all of the above methods are limited to using offline features for uplift modeling and identifying sensitive target population, which means they fail to capture users' real-time interests, thereby being unable to dynamically adjust the target population with specific treatment. Moreover, the design and utilization of treatments in these methods cannot be directly transferred to the context of video recommendations.

Some recent works have also explored the user feedback modeling methods in online video recommendation. MIKD [41] captures the dynamic dependence between context cumulative effects and user feedback through knowledge distillation; CWM [46] offsets duration bias through counterfactual modeling; and [21] uses user feedback for diverse recommendations on full-stage online video platforms. Our proposed CDUM overcomes the above shortcomings and designs a treatment plan for the video recommendation scenario. It initially models users' long-term preferences at a coarse granularity using daily-level features, then refines the uplift modeling by capturing users' real-time interests through online real-time features, dynamically adjusting the target population for the specific treatment.

3 PRELIMINARIES

Since our designed model consists of both offline and online components, in this section, we will define and explain the problem formulation for both offline and online scenarios separately.

3.1 Offline Problem Definition

We denote $\mathcal{Z} = \{\mathbf{z}_1, \mathbf{z}_2, ..., \mathbf{z}_N\}$ as the set of instances at the video recommendation scenarios, where $\mathbf{z}_i = (\mathbf{x}_i, \mathbf{t}_i, y_i) \in \mathcal{X} \times \mathcal{T} \times \mathcal{Y}$ and N is the number of training instances. $\mathbf{x}_i \in \mathbb{R}^{f_x \times 1}$ represents the f_x-dimensional user features of the i-th instance. While $\mathbf{t}_i \in \mathbb{R}^{f_t \times 1}$ is the f_t-dimensional treatment features for the i-th instance. Besides, $y_i \in \{y_i^0, y_i^1, ..., y_i^K\}$ is the response label under specific treatment for the i-th instance. Besides, we consider the first treatment feature

of each instance to represent the treatment's index ID, with a total of K treatments, i.e., $t_i^0 \in [0, K]$.

Following the Neyman-Rubin potential outcome framework [31], we denote y_i^k and y_i^0 as the potential outcome when the user in the i-th instance gets a particular treatment $t_i^0 \in [1, K]$ or is not treated, respectively. As we can not observe y_i^k and y_i^0 at the same time, there is no true uplift result for each instance, which is known as a key reason uplift modeling differs from traditional supervised learning. Thus, we need uplift modeling to estimate the expected individual treatment effect $\tau_k(x_i)$ for each instance.

Different from the existing uplift methods which estimates $\tau_k(x_i)$ with the calculation of the difference between $\mathbb{E}(y_i^k|t_i^0 = k, x_i)$ and $\mathbb{E}(y_i^0|t_i^0 = 0, x_i)$, we pass $\mathbb{E}(y_i^k|t_i^0 = k, x_i)$ and $\mathbb{E}(y_i^0|t_i^0 = 0, x_i)$ into online and combine online module (CPM) to estimate $\tau_k(x_i)$. As a result, we can dynamically refine the estimation of ITE scores and adjust the effective users under different treatments in real-time.

3.2 Online Problem Definition

In the online video recommendation pipeline, we have real-time contextual features (i.e, video features of several recent user interactions) and request-level candidates (i.e, all candidate video features under a single user request), which are represented as $\mathbf{s}_i = [\mathbf{s}_i^0, \mathbf{s}_i^1, ..., \mathbf{s}_i^{f_s}]$. Here $\mathbf{s}_i^j \in \mathbb{R}^{L^j \times 1}$ represents the j-th contextual features or request-level candidates for the i-th instance, and L^j represents the number of recent videos with which the user has interacted. For instance, in response to a user's request, the online recommendation pipeline generates L^1 candidate videos, and at this time we intercept L^2 videos of the user's most recent interaction. These videos have rich features (such as pctr, a click-through estimate, which is a estimate between 0 and 1). It is worth noticing that we leverage a request-level instance to train the online module, i.e., modeling an instance using the features and label of each user request. On this basis, we further design $r_i^k \in \{r_i^1, ...r_i^K\}$ as the request-level label of online module. Here K corresponds to the number of treatments. To model the real-time interest of the users dynamically, r_i^k is designed to be calculated from the interaction data of exposed videos within the user candidate set at the request-level. Specifically, we divide the exposed videos into K categories based on their durations, which directly corresponds to the design of our treatment. Next, for the k-th duration category, we calculate the ratio of the number of long-play to short-play videos for each user and denote it as r_i^k. Additionally, we use $\hat{r}_i^k$ as the real-time interest score, representing the user's preference for videos of different durations. This score is used to adjust the offline long-term preference score $\mathbb{E}(y_i^k|t_i^0 = k, x_i)$, allowing us to estimate the user's individual treatment effect scores. Finally, we can rank leverage this score to make a rational treatment assignment.

4 METHOD

We propose Coarse-to-fine Dynamic Uplift Modeling (CDUM) for Real-time Video Recommendation, which includes two modules: Coarse-grained Preference Modeling (CPM) and Fine-grained Interest Capture (FIC). Figure 2 illustrates the technical details of the proposed model.

4.1 Coarse-grained Preference Modeling

In this part, we introduce a module, Coarse-grained Preference Modeling, which utilizes the offline features to model the user's long-term preferences for different treatments.

4.1.1 Feature Encoder Module. For each instance $\mathbf{z}_i = (\mathbf{x}_i, \mathbf{t}_i, y_i)$, we encode both $\mathbf{x}_i$ and $\mathbf{t}_i$ into embeddings. For the continuous feature, we first utilize a equidistant bucketing approach to convert continuous features into sparse features. Then, we initialize embedding tables for both processed continuous and sparse features, and capture the embedding representations corresponding to the specific feature value. Specifically, we have:

$$\mathbf{e}_{xi} = \mathbf{E}_x^T \cdot \mathbf{x}_i, \ \mathbf{e}_{ti} = \mathbf{E}_t^T \cdot \mathbf{t}_i, \tag{1}$$

where $\mathbf{E}_x \in \mathbb{R}^{f_x \times d}$ and $\mathbf{E}_t \in \mathbb{R}^{f_t \times d}$ are the created embedding tables, f_x and f_t are the numbers of non-treatment features and treatment features, and d is the embedding size. In addition, we can get the corresponding embeddings for each feature, i.e., $\mathbf{e}_{xi} = \{\mathbf{e}_{xi}^0, \mathbf{e}_{xi}^1, ..., \mathbf{e}_{xi}^{f_x-1}\}$ and $\mathbf{e}_{ti} = \{\mathbf{e}_{ti}^0, \mathbf{e}_{ti}^1, ..., \mathbf{e}_{ti}^{f_t-1}\}$

4.1.2 Causal Effects Learning Module. As we propose to *manage the distribution of videos with varying durations*, we correspond each treatment to a specific adjustment of the video type distribution, i.e., learning the preference of users (expressed as responses) for different treatments. First, we encode the non-treatment embeddings of users with a concatenate operation, and use an average pooling operation on the treatment embedding:

$$\mathbf{e}_{xi}^* = \underset{j=0}{\overset{f_x}{Concat}}(\mathbf{e}_{xi}^j), \mathbf{e}_{ti}^* = \sum_{j=0}^{f_t} \mathbf{e}_{ti}^j / (f_t + 1), \tag{2}$$

Additionally, to fully leverage the treatment information, inspired by multi-interest learning [5, 20, 28], we further refine treatment embedding into indicator embedding and guidance embedding. We consider these two forms of embeddings as interest generalization under different effects of treatment. For the guidance part, we expect it to serve as a filter and information extractor for user features, expressing the guiding role of the treatment. In contrast, for the indicator part, we expect it to express the indicative role of the treatment, thereby helping the model generalize to learning tasks involving different treatments. Specifically, we first utilize an interest extraction module to refine the guidance and indicator parts from $\mathbf{e}_{ti}$. Take the guidance part as an example, we have:

$$\begin{cases} \mathbf{h}_{ti}^{gui} = ReLU\left(\mathbf{e}_{ti}^* \cdot \mathbf{W}_{gui}^h + \mathbf{b}_{gui}^h\right), \\ \mathbf{e}_{ti}^{gui} = Sigmoid\left(\mathbf{h}_{ti}^{gui} \cdot \mathbf{W}_{gui}^o + \mathbf{b}_{gui}^o\right), \end{cases} \tag{3}$$

where $\mathbf{W}_{gui}^h \in \mathbb{R}^{d \times d_h}$ and $\mathbf{b}_{gui}^h \in \mathbb{R}^{1 \times d_h}$ are transformation matrix and bias matrix for the hidden layer, d_h is the dimension of the hidden representation $\mathbf{h}_{ti}^{gui}$. $\mathbf{W}_{gui}^o \in \mathbb{R}^{d_h \times d^*}$ and $\mathbf{b}_{gui}^o \in \mathbb{R}^{1 \times d^*}$ are transformation matrix and bias matrix for the output layer. $\mathbf{e}_{ti}^{gui}$ is the guidance embedding refined from $\mathbf{e}_{ti}$, and d^* is the dimension of the embedding. While $\mathbf{e}_{ti}^{ind}$ which represents the indicator embedding can be obtained in a similar process.

Furthermore, we first apply operation to $\mathbf{e}_{xi}^*$ similar as Equation (3), and obtain M expert representations (denoted as $\mathbf{f}_m$, $m \in \{1, 2, ..., M\}$). After that, we utilize the guidance embedding to filter

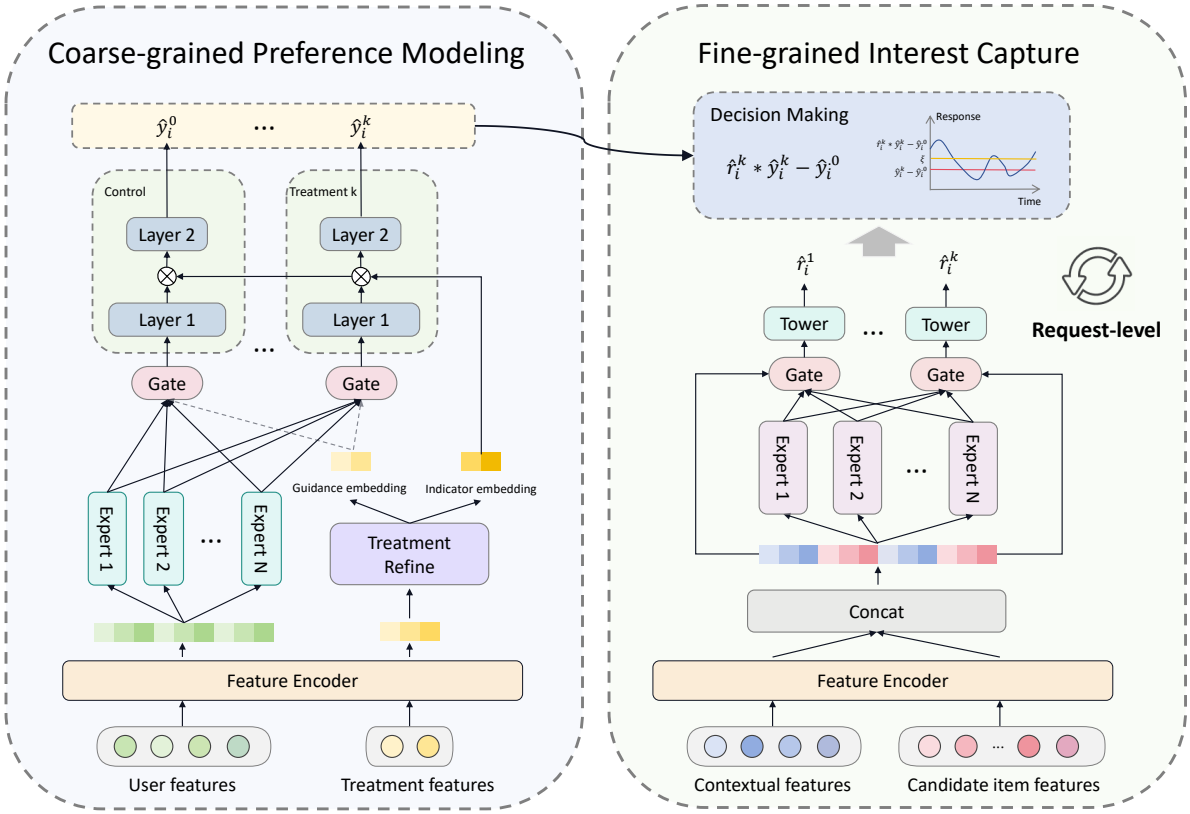


Figure 2: Illustration of the proposed CDUM framework. The light purple block on the left is the offline training part (CPM), and the light green block on the right is the online inference part (FIC).

and extractor information for user features while leveraging the indicator embedding to enhance the representations of the extracted information, and have:

$$\begin{cases} \mathbf{g}_i^k = Softmax(\mathbf{W}_g \cdot \mathbf{e}_{ti}^{gui} + \mathbf{b}_g) \\ \hat{y}_i^k = h^k\left(\mathbf{e}_{ti}^{ind}, \sum_{m=1}^{M} \mathbf{g}_i^k(m) \cdot \mathbf{f}_m\right) \end{cases} \tag{4}$$

where $\mathbf{W}_g \in \mathbb{R}^{M \times d^*}$ and $\mathbf{b}_g \in \mathbb{R}^{M \times 1}$ are feature transformation matrix and bias matrix, and $\mathbf{g}_i^k \in \mathbb{R}^{M \times 1}$ is the attention vector which is used as selector to calculate the weighted sum of all experts. $\mathbf{g}_i^k(m)$ denotes the m-th element of vector $\mathbf{g}_i^k$, $h^k(\cdot)$ is the tower function, which typically consists of several fully connected layers used to model high-level interactions for the k-th treatment. In addition, $\mathbf{e}_{ti}^{ind}$ is applied as the form of mask before the final output of the tower. $\hat{y}_i^k$ denotes the prediction score for the response of user under the k-th treatment.

4.1.3 Training Objective and Prediction. As we have obtained $\hat{y}_i^k$ for the response of user under k-th treatment, we further use the huber loss [15] as the training loss function, i.e.,

$$\mathcal{L} = \begin{cases} 0.5 * (\hat{y}_i^k - y_i^k)^2, & if |\hat{y}_i^k - y_i^k| \le \delta \\ 0.5 * \delta^2 + \delta * (|\hat{y}_i^k - y_i^k| - \delta), & if |\hat{y}_i^k - y_i^k| > \delta \end{cases} \tag{5}$$

where y_i^k is the response label under k-th treatment. δ is a constant that can be adjusted dynamically. It is worth noting that in the training stage, since each sample $(\mathbf{x}_i, \mathbf{t}_i, y_i)$ is obtained by recording response only under the action of a specific treatment (e.g, k-th treatment), we only use the output $\hat{y}_i^k$ of tower corresponding to the treatment for training, while mask operation is applied to the calculation of loss to other towers.

Finally, at the prediction stage, we can infer $\hat{y}_i^k (k \in \{0, 1, 2, ..., K\})$ for each user by replacing the feature of treatment $\mathbf{t}_i$ iteratively. Further, we can obtain the long-term preference scores of users for different treatments, and deliver them to the next online part, FIC.

4.2 Fine-grained Interest Capture

Although we have captured the long-term preference scores, the targeted decision of the affected group still can not be adjusted dynamically in real time. In the pipeline of online video recommendation, there is rich and varied feature information, such as online real-time contextual features and request-level candidates information. Thus, we innovatively design a module, Fine-grained Interest Capture Module. This module can leverage the online features to model users' real-time interests and dynamically adjust long-term preferences accordingly, enabling real-time dynamic adjustment of the effective user groups for different treatments.

4.2.1 Module Structue. For the selection of features and labels, as mentioned in Section 3.2, we use contextual features and request-level candidates as features, i.e., $\mathbf{s}_i$. At the same time, the ratio of the number of long and short broadcast exposed videos of users under each treatment classification (divided by different durations of videos) is used as the label of corresponding categories.

We first encode the online features to embeddings. This process is similar to Section 4.1.1, thus we get the online embeddings $\mathbf{e}_{si} = \{\mathbf{e}^0_{si}, \mathbf{e}^1_{si}, ..., \mathbf{e}^{f_s}_{si}\}$. Here $\mathbf{e}^j_{si} \in \mathbb{R}^{1\times L^j\times d_s}, j \in [0, f_s]$, L^j represents the number of recent videos with which the user has interacted, d_s is the embedding size. Then, we apply an average pooling to the second dimension, and concatenate all the pooled embeddings together. This process can be represented as:

$$\mathbf{e}^*_{si} = \underset{j=0}{\overset{f_s}{Concat}}\left(\sum_{l=1}^{L^j} \mathbf{e}^j_{si}(l)/(L^j)\right), \tag{6}$$

where $\mathbf{e}^j_{si}(l) \in \mathbb{R}^{1\times 1\times d_s}$ denotes the l-th interacted item embedding.

Further, we equip $\mathbf{e}^*_{si}$ with a multi-task learning network. Here, the form of the network is in the form of a multi-gate mixture-of-experts module [26, 27], and for simplify, we denote $\hat{r}^k_i$ ($k \in [1, K]$) as the output of each tower of this part.

4.2.2 Training Objective and Prediction. In the previous section, we have obtained the output $\hat{r}^k_i$ for real-time interest learning of different treatments. As mentioned in Section 4.1.1, we design an online request-level label, which is the ratio of the number of long-playing videos to the number of short-playing videos. To be specific, suppose that the number of exposed videos for a user's request is V and we have K categories of treatment. The designation of the label can be formulated as:

$$\sum_{k=1}^{K} V_k = V, V^{long}_k + V^{short}_k = V_k, r^k_i = \frac{V^{long}_k + 1}{V^{short}_k + 1}, \tag{7}$$

where V_k denotes the k-th category of video duration (i.e., treatment). V^{long}_k and V^{short}_k represent the numbers of long-playing and short-playing videos, respectively. Here, we'll just use long- and short-playing videos as an example. In the actual deployment, we use the posterior statistics of the all videos duration to carry out equal frequency buckets to divide the video categories of different lengths.

Furthermore, for the training, we utilize the huber loss as optimization function, which is similar to the process in Section 4.1.3. While for the prediction, we deploy this module directly online, and it is invoked with every user request to calculate the real-time interest score $\hat{r}^k_i$.

4.2.3 Decision Making. As we have obtained the offline long-term preference score $\hat{y}^k_i (k \in \{0, 1, ..., K\})$ and online real-time interest score $\hat{r}^k_i (k \in \{1, 2, ..., K\})$, we can eventually combine them to dynamically adjust the effective user groups for different treatments. Specifically, as shown in Figure 2, we define $\hat{r}^k_i * \hat{y}^k_i - \hat{y}^0_i$ as the decision score for the k-th treatment. We then define ξ as the threshold for whether the treatment takes effect (i.e., if $\hat{r}^k_i * \hat{y}^k_i - \hat{y}^0_i > \xi$, the k-th treatment is enabled). Thus, the treatments (i.e., adjusting the distribution of videos with varying durations) for each user will be dynamically adjusted to enable or disable in real time. A detailed overview can be found in Section 6.2.

5 OFFLINE EVALUATIONS

5.1 Experimental Setting

Table 1: Statistics of the two public datasets

Dataset	LAZADA		CRITEO
Type	Train	Test	All
Size	926,669	181,669	13,979,592
Ratio of Treatment to Control	0.28:1	1.09:1	5.67:1
Positive Sample Ratio	1.99%	3.52%	4.70%
Features Number	83		12
Conversion Target	Label		Visit

5.1.1 Dataset Description. We evaluate our proposed method on both public and real-world industrial datasets. For the public dataset, we use CRITEO [11] and LAZADA [48]. CRITEO is a dataset built by Criteo AI Labs in a real large-scale advertising scenario. It includes nearly 14 million instances, 12 features, a treatment indicator, and two labels (i.e., visit and conversion). We use visit as the target, and randomly split the dataset for the training, validation, and testing with a ratio of 8:1:1. LAZADA is a real voucher distribution business scenario dataset from the e-commerce platform Lazada. It contains 83 features, a treatment indicator, and a label. Due to the operational target strategy, the treatment allocation is selective in the actual production environment, these data containing treatment bias are utilized as training set, and a small number of other users who are not affected by the targeting strategy are used as testing set, and their treatment allocation follows the randomized controlled trial. The statistical information of the two datasets is summarized in Table 1.

The industrial dataset consists of online user data collected during the 14 days leading up to the start of multiple randomized controlled trials (RCTs), along with user feedback data gathered throughout the trials. To be specific, we conducted online experiments with several random population groups, where one of the group is non-treated (control) and others received different treatments (increasing the proportion of videos with different durations)). To ensure fairness and eliminate bias, we applied a reshuffling technique before the experiment to balance these groups in terms of key historical metrics (e.g., watch time, play count) over the past 7 days. This ensures that the collected RCT data is unbiased and avoids issues such as content exposure bias or user-level unfairness. The dataset includes a variety of features, such as users' inherent attributes (e.g., region, activity level), consumption behavior over the past 7 and 30 days (e.g., app usage duration, video views), and commercial attributes (e.g., user commercial value classification). Furthermore, we computed the *average app usage time* during the experiment period to serve as the target label. Each treatment group contains approximately 8,500,000 instances. The dataset was divided into training, validation, and test sets in an 8:1:1 ratio.

5.1.2 Evaluation Metrics. In our experiments, we adopt three widely used metrics to evaluate uplift ranking performance of different models: normalized Area Under the Uplift Curve (AUUC), normalized area under the qini curve (QINI), and uplift score at first h percentile (LIFT@h, h is set as 30). We compute these metrics using a standard python package scikit-uplift[1].

5.1.3 Baseline Models. To demonstrate the effectiveness of our proposed CDUM, we selected a set of currently popular and representative uplift modeling methods, including:

- **S-Learner** [19]: A classic meta-learning method that estimates the uplift of all treatments through a single model.
- **T-Learner** [19]: An improved meta-learning method based on S-Learner. Specifically, it estimates the treatment group and the control group separately by setting up an additional model, and obtains the specific ITE through the difference between this two.
- **BNN** [16]: It introduces deep neural networks into gain modeling and proposes a new algorithmic framework for counterfactual inference.
- **TARNet** [32]: It is a variant of CFRNet, with the balancing regularization component removed.
- **CFRNet** [32]: This method aligns treated and control distributions, and present a novel generalization-error bound that connects ITE estimation error with standard generalization-error and distributional distance. Explicit bounds for Wasserstein and Maximum Mean Discrepancy (MMD) distances are derived using Integral Probability Metrics.
- **CEVAE** [25]: The method focuses on handling and modeling confounding factors by proposing a causal modeling approach based on variational autoencoders (VAE), achieving more robust results.
- **GANITE** [43]: A method based on Generative Adversarial Networks (GAN) is proposed for inferring individual treatment effects. Specifically, the approach employs GAN framework to construct a counterfactual generator and an ITE generator. The counterfactual generator is used to learn unseen counterfactuals, while the ITE generator is trained based on these to derive the individual treatment effects.
- **DragonNet** [33]: A neural network-based approach that uses propensity scores for estimation adjustment, and targeted regularization is applied to the model to enhance the quality of the final treatment effect estimation.
- **FlexTENet** [7]: An architecture based on multi-task learning incorporates the idea of adaptively learning shared content when estimating an individual's potential outcomes, aimed at exploring the conditional average treatment effect (CATE) problem for binary treatments.
- **SNet** [6]: Based on the shortcomings of previous neural network-based methods, it proposes a new general network architecture for learning multiple representations, allowing potential outcome regression to depend only on a subset of shared features.
- **EUEN** [17]: An end-to-end uplift modeling approach that uses different networks to separately model the control conversion probability and the uplift effect of the treatment.
- **DESCN** [48]: This method employs a multi-task learning framework to jointly model treatment and response functions. It introduces a cross-network to learn integrated information about the treatment while addressing treatment bias and sample imbalance issues.
- **EFIN** [22]: A method for feature extraction through feature interaction. It focuses on the sensitivity of non-treatment features to specific treatments and enhances model robustness using a causal constraint module.

For the experiments on the industrial dataset, we utilize some classic multi-task models to replace the CPM model in order to test the effectiveness, including Shared Bottom [4], MMOE [26], PLE [37], MESI [12]. Since the dataset contains multiple treatments, the final evaluation metric is the average of the AUUC scores of all treatments, named AUUC_{avg}.

5.1.4 Parameter Settings. Our proposed CDUM is implemented in TensorFlow [1]. We use the Adam optimizer [18] with a learning rate of 0.001 and set the learning rate reduction factor to 0.6. The number of allowed epochs with no improvement is 2, after which the learning rate will be reduced. In addition, we set the batch size to 4096 and the embedding dimension to 32. The number of expert modules in CPM is 3. The default values of δ and ξ are 1 and 0, respectively.

Table 2: The overall performance comparison on Lazada and Criteo datasets. Boldface denotes the highest score and underline indicates the results of the best baselines.

Dataset	LAZADA			CRITEO		
Metrics	QINI	AUUC	LIFT@30	QINI	AUUC	LIFT@30
S-Learner	0.0207	0.0029	0.0072	0.0902	0.0354	0.0298
T-Learner	0.0217	0.0031	0.0084	0.0790	0.0310	0.0277
BNN	0.0230	0.0033	0.0073	0.0888	0.0349	0.0286
TARNet	0.0214	0.0030	0.0072	0.0847	0.0333	0.0288
CFRNet	0.0231	0.0033	<u>0.0087</u>	0.0901	0.0354	<u>0.0302</u>
CEVAE	0.0158	0.0022	0.0074	0.0867	0.0341	0.0290
GANITE	0.0174	0.0024	0.0078	0.0818	0.0322	0.0283
DragonNet	0.0176	0.0025	0.0077	0.0851	0.0335	0.0289
FlexTENet	0.0185	0.0026	0.0079	<u>0.0924</u>	<u>0.0363</u>	0.0298
SNet	<u>0.0238</u>	<u>0.0034</u>	0.0076	0.0843	0.0331	0.0284
EUEN	0.0156	0.0022	0.0073	0.0898	0.0353	0.0300
DESCN	0.0223	0.0031	0.0075	0.0803	0.0316	0.0274
EFIN	0.0141	0.0020	0.0077	0.0859	0.0337	0.0287
CPM	**0.0265**	**0.0037**	**0.0087**	**0.0942**	**0.0371**	**0.0310**

5.2 Performance Comparison

We report the evaluation results of the offline module CPM in our proposed CDUM and the baselines on two offline public datasets in Table 2. From the table, we can see that:

- On both two datasets, the performance of S-learner and T-Learner is not bad and even outperforms some baselines using more complex network architectures. This shows that in the scenario of large-scale uplift modeling, complex modeling methods may lead to suboptimal result due to the inability to quickly identify the sensitive features of users.

[1]https://www.uplift-modeling.com/en/latest

Table 3: The overall performance comparison on industrial offline dataset. Boldface denotes the highest score and underline indicates the results of the best baselines.

Dataset	Industrial Dataset
Metrics	$AUUC_{avg}$
S-learner	0.00876
T-Learner	0.00787
CausalForestDML	0.00914
EFIN	0.00841
Shared Bottom	0.00798
MMOE	0.00947
PLE	0.00998
MESI	<u>0.01053</u>
CPM	**0.01143**

Table 4: Performance of different CPM variants. Boldface denotes the highest score.

Dataset	LAZADA			CRITEO		
Metrics	QINI	AUUC	LIFT@30	QINI	AUUC	LIFT@30
CPM *w/o* indicator	0.0218	0.0030	0.0081	0.0922	0.0362	0.0308
CPM *w/o* guidance	0.0220	0.0031	0.0081	0.0901	0.0354	0.0304
CPM	**0.0265**	**0.0037**	**0.0087**	**0.0942**	**0.0371**	**0.0310**

- On the LAZADA dataset, SNet performs better than other baselines overall, but on the CRITEO dataset, the best performing baseline is FlexTENet. In addition, some baselines (e.g., EUEN and DESCN) exhibit significant performance variations when evaluate on the two datasets. This inconsistency highlights that current baseline models are unable to provide reliable performance across different scenarios, lacking strong generalizability.
- Our proposed CPM performs best in all metrics of the two datasets, which demonstrates the effectiveness and applicability of our proposed offline module CPM in the task of uplift modeling.

The problem of managing the distribution of videos with varying durations can also be addressed using a multi-task learning framework. To further validate the performance of CPM and to integrate it with the online module FIC for deployment, we conducted experiments on our industrial dataset. Table 3 shows that CPM still maintains superior performance in comparison with classic multi-task learning models.

5.3 Ablation Study

In order to evaluate the effectiveness of the key design of the offline module CPM in our proposed method CDUM, we conduct ablation studies. Specifically, we design the following two variants:

- CPM w/o indicator: We remove the indicator embedding used to enhance the representation of extracted information.
- CPM w/o guidance: We use the initial feature embedding instead of the guidance embedding to generate the weights when aggregating experts in gating.

The results are shown in Table 4. It can be seen that removing any design will cause performance degradation. This demonstrates the rationality and effectiveness of our proposed module.

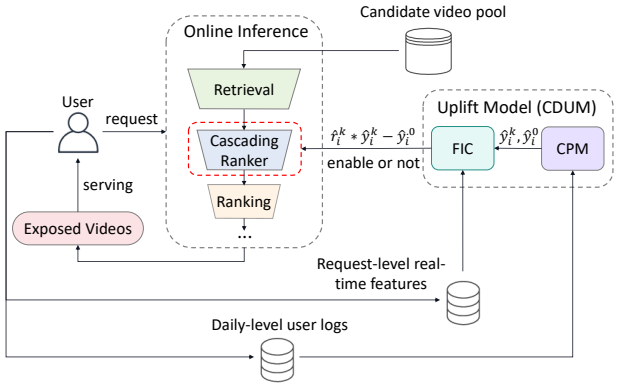


Figure 3: Illustration of the online pipeline.

6 ONLINE APPLICATION

To further evaluate the online performance of our model in real product scenarios, we apply CDUM by adding a new model in the industrial recommendation service of Kuaishou platform, which serves hundreds of millions of users. We then deploy them online for 18 days of online A/B testing.

6.1 System Description

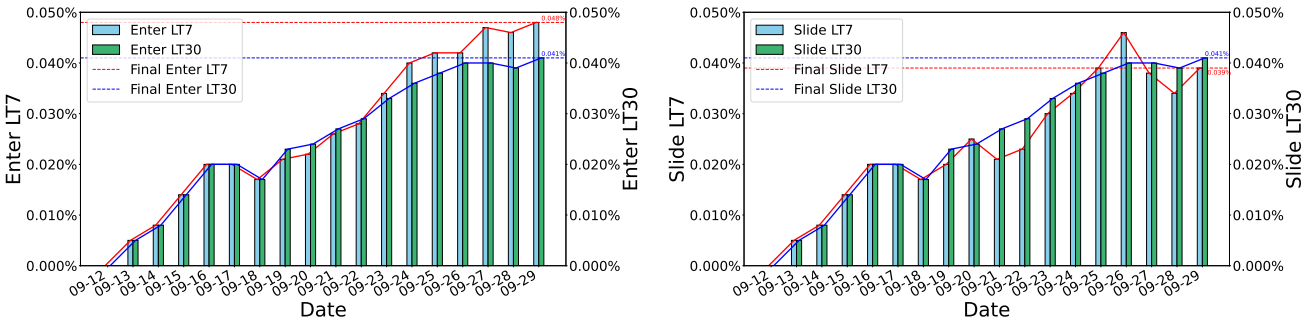
As shown in Figure 3, in the pipeline of online video recommendation, a series of modules are cascaded together to filter candidate videos and finally expose the selected videos to the user. Specifically, when a user triggers a request, the corresponding user's attributes and contextual features are sent to the online service. And the online pipeline is triggered to filter and select the videos from the candidate pool in a cascading form. At last, the selected videos are exposed to the user, thus the request is over.

6.2 Scheme Overview

Our model, as illustrated in Figure 3, is developed at the bucket video duration adjustment part of *Cascading Ranker*. To be specific, when a user sends a request, the retrieval part first retrieves candidate videos from a candidate pool, and sends them to the Cascading Ranker. Therefore, the Cascading Ranker is triggered, and in this stage, CDUM is invoked to inference with the input of user request-level features and user's attributes, etc. In CDUM, the outputs $\hat{y}_i^k, \hat{y}_i^0$ of CPM are updated at a daily level, thus they are directly utilized to be calculated for the decision scores. While the FIC module makes inference based on the incoming features with the user request in real time and outputs $\hat{r}_i^k$. These two parts of CDUM work together to decide the enablement of treatments at the level of the user's request, thus directly acting at the bucket video duration adjustment part of the Cascading Ranker.

6.3 Online Experimental Results

6.3.1 Online A/B test results. We leverage the common consumption metrics (i.e., APP usage time and watch time) and retention metrics (i.e., Next day retention, 7-day retention, Enter LT7/LT30 and Slide LT7/LT30) in the industry to evaluate the online performance. Among them, **LT7/LT30** is an unbiased estimate of medium-



(a) Enter LT (b) Slide LT

Figure 4: Online A/B test experimental results of Enter LT and Slide LT.

and long-term DAU (Daily Active Users) for experimental evaluation, pointing strictly to user experience. They are key metrics that indicate user retention benefits online, and can be formulated as:

$$\begin{cases} \text{Enter LT7} = \dfrac{\sum_{i=max(T-6,T_0)}^{T} DAU_i}{\text{Total number of active users from } T_0 \text{ to } T} \\ \text{Slide LT7} = \dfrac{\sum_{i=max(T-6,T_0)}^{T} DAU_i}{WAU} \end{cases} \tag{8}$$

where T stands for the date the experiment data is recorded, T_0 denotes the date that the experiment is started. DAU_i represents the number of daily active users at the i-th day. WAU represents the number of weekly active users calculated from $max(T-6, T_0)$ to T. The LT30 is calculated in a similar way. And as the platform serves hundreds of millions of users every day, it is known that a slightly higher LT7/LT30 (↑) at ***0.01%-level*** is regarded significant.

Figure 4 shows the performance of our model in improving the LT metrics during the test period. We can find that our model achieves significant improvements of **0.048%**/**0.041%** and **0.041%**/**0.039%** with respect to Enter LT7/LT30 and Silde LT7/LT30 respectively, which verifies the effectiveness of our proposed method. Besides, as shown in Table 5, our model shows significant improvement over baseline in other retention and consumption metrics, again demonstrating the superiority of CDUM. Meanwhile, the result of the online latency comparison indicates that our model achieves comparable efficiency with negligible overhead, making it suitable for real-time deployment. Last but not least, our model is now deployed online with full users at Kuaishou platform, serving for hundreds of millions of users every day.

6.3.2 Ablation Test of CDUM. Furthermore, to demonstrate the effectiveness of the online module FIC, we conduct an ablation experiment for our proposed CDUM. To be specific, we remove the decision-making part of online adjustment, and rely solely on $\hat{y}_i^k - \hat{y}_i^0 > \xi$ ($k \in [1, K]$) to determine whether the k-th treatment is enabled or not.

As illustrated in Table 6, CDUM significantly outperforms its variant CDUM w/o FIC on consumption metrics, proving the indispensability of FIC.

7 CONCLUSIONS AND FUTURE WORK

In this paper, we explore uplift modeling in the context of video recommendation. To appropriately design treatments that capture user's real-time interests while watching videos, we propose Coarse-to-fine Dynamic Uplift Modeling (CDUM) for real-time video recommendation. This model leverages a Coarse-grained Preference Modeling (CPM) module to learn users' long-term preferences using their offline daily-level features, and then employs a Fine-grained Interest Capture (FIC) module for online inference based on real-time features, thereby capturing users' immediate interests. Extensive offline experiments on public and industrial datasets as well as online A/B test on Kuaishou platform have demonstrated the effectiveness and superiority of our method.

For future work, we plan to explore the potential benefits of enhancing modeling at other stages of the recommendation pipeline, such as ranking, while further designing and improving the model structure and the utilization of features.

Table 5: The performance of Online A/B Testing at the platform. Boldface represents significance level p-value < 0.05 of comparing CDUM with the online baseline.

Dataset	Online Platform	
Metrics	Online Baseline	**CDUM**
Next day retention	-	**+0.034%**
7-day retention	-	**+0.038%**
APP usage time	-	**+0.135%**
APP usage time (per capita)	-	**+0.131%**
watch time	-	**+0.188%**
Latency	-	+0.009%

Table 6: The ablation result of Online A/B Testing at the platform. Boldface represents significance level p-value < 0.05 of comparing CDUM with the variant.

Scenario	Online Platform	
Metrics	CDUM w/o FIC	**CDUM**
APP usage time	-	**+0.078%**
APP usage time (per capita)	-	**+0.094%**
watch time	-	**+0.129%**

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